Mediapipe Human Pose Tracking (PTZ)

1. Experiment Objective

The PTZ pan-tilt tracks the human body position. Based on MediaPipe Pose, it detects 33 whole-body landmarks (including face, hands, and feet), uses the average center of all landmarks as the tracking target, and drives the pan-tilt with PD control so that the human body stays at the center of the frame.

2. Experiment Principle

1. Key Terms

Term	Description
MediaPipe Pose	Google's open-source human pose estimation framework; detects 33 whole-body landmarks (including face, hands, and feet)
Pose Landmark Average Center	The arithmetic mean of the coordinates of all detected landmarks, used as an estimate of the overall human body position
Visual Tracking	Detects and continuously locks onto a target through a vision algorithm; only the pan-tilt follows, the car does not move
PTZ Pan-Tilt	A two-degree-of-freedom servo pan-tilt; s1 rotates horizontally and s2 tilts, controlled via the `/cmd_ptz_angle` topic
PD Control	Proportional-derivative control; computes the pan-tilt angle step from the target offset and the offset change rate
Deadband	When the target offset is within the deadband pixel range, the pan-tilt does not move, avoiding continuous jitter caused by tiny errors

2. Technical Architecture

```
State estimation (ekf_filter_node + camera_node) → pose_tracker_ptz_node →
/cmd_ptz_angle → firmware servo control
```

Node chain explanation:

- **ekf_filter_node**: Extended Kalman filter node that fuses IMU/odometry data for state estimation
- **camera_node**: Camera driver node that handles the image topic published by the ESP32
- **pose_tracker_ptz_node**: MediaPipe human pose tracking node that detects 33 landmarks, computes the average center, and calculates the pan-tilt angle step with a PD controller
- **/cmd_ptz_angle**: PTZ servo angle command topic, bridged to the ESP32 firmware by micro_ros_agent, controlling the s1 (horizontal) and s2 (tilt) servos

3. Core Topics Quick Reference

Firmware Uplink (ESP32 firmware → micro_ros_agent → ROS2 node)

Topic Name	Type	QoS	Description
`/camera/image_raw`	`sensor_msgs/Image`	BEST_EFFORT, KEEP_LAST, depth=1, VOLATILE	Raw image frames published by the ESP32 camera

Firmware Downlink (ROS2 node → micro_ros_agent → ESP32 firmware)

Topic Name	Type	QoS	Description
`/cmd_ptz_angle`	`geometry_msgs/Vector3`	RELIABLE, KEEP_LAST, depth=10	PTZ pan-tilt angle command (s1: horizontal angle, s2: tilt angle)

ROS2 Internal (inter-node communication)

Topic Name	Type	QoS	Description
`/camera/image_raw`	`sensor_msgs/Image`	BEST_EFFORT	Camera image (bridged to ROS2 by micro_ros_agent; subscribed by the tracking node)
`/cmd_ptz_angle`	`geometry_msgs/Vector3`	RELIABLE	PTZ angle command (published by the tracking node; bridged to the firmware by micro_ros_agent)

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑
No-PTZ version	❌
Required peripherals	ESP32 camera (pan-tilt) + PTZ servo

2. Start Agent + Camera + Joint Model (Terminal 1)

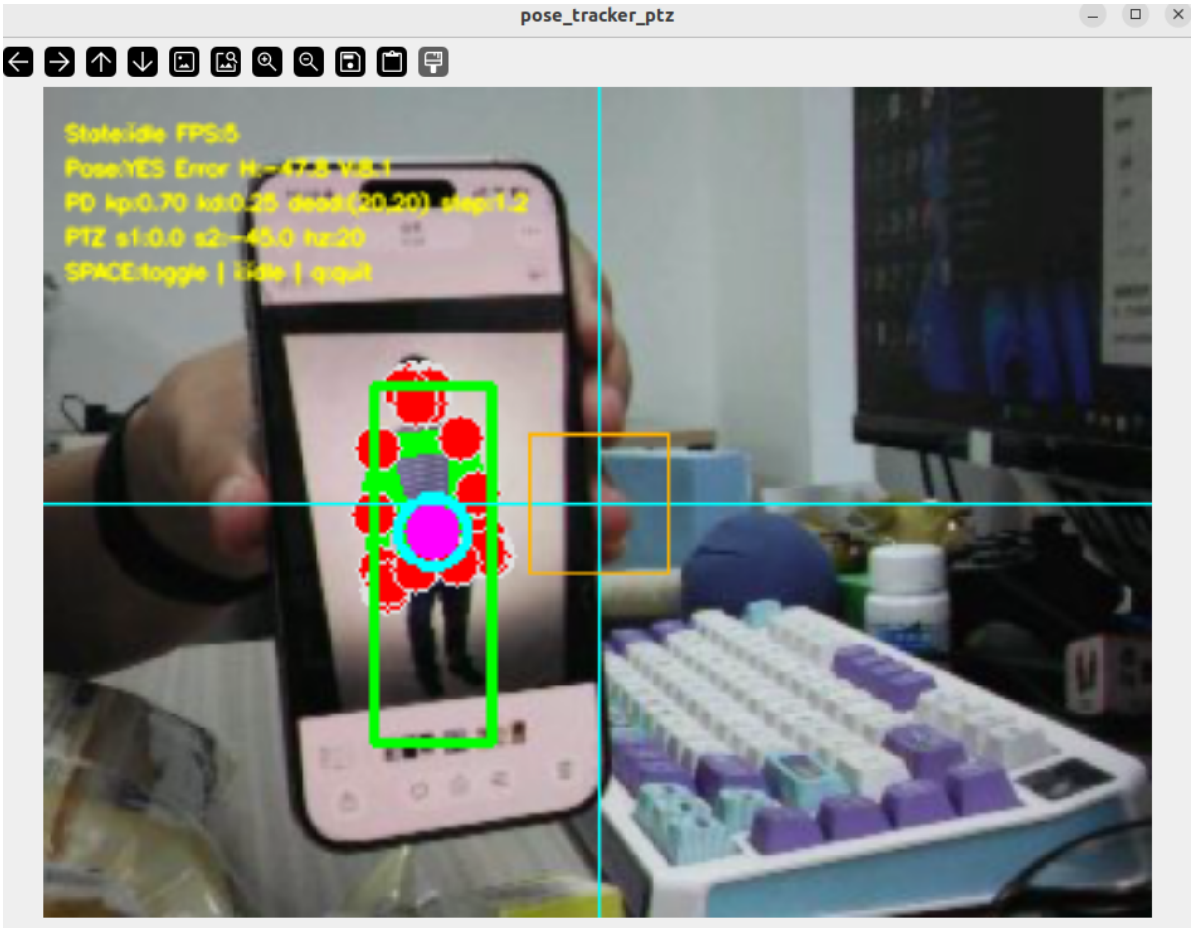
```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info     | UDPv4AgentLinux.cpp | init                  | running...            | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start Mediapipe Human Pose Tracking (Terminal 2)

Open a second terminal, activate the MediaPipe environment, and start the human pose tracking node:

```
ros2 launch microros_v2_ptz_visual_control_pkg pose_tracker_ptz.launch.py
```



4. Operation Instructions

After startup, the node uses two independent timers: image processing (`image_processing_rate_hz` = 15 Hz) + pan-tilt control (`control_rate_hz` = 20 Hz). Minimal two-state machine: `idle` ↔ `tracking`.

Image reception and preprocessing

Subscribes to `/camera/image_raw` (QoS: BEST_EFFORT, KEEP_LAST); the image callback converts and caches the raw Image. The image processing timer takes a copy of the latest frame, scales it to the processing resolution (`processing_width` × `processing_height` = 320×240), and flips/rotates as needed.

MediaPipe Pose detection

Uses `PoseDetector` (based on MediaPipe Pose, detection confidence = `pose_detection_confidence` = 0.5, tracking confidence = `pose_tracking_confidence` = 0.5) to detect the human pose. `pubPosePoint()` returns the drawn image frame, the list of 33 landmarks, and the bbox information.

Tracking target computation

Takes the mean of the x coordinates and the mean of the y coordinates of all detected 33 landmarks as the overall body center (cx, cy). This algorithm is more stable than a single landmark (such as the nose tip): even if some landmarks are occluded or leave the frame, a good position estimate is still maintained.

Pose visualization

On-screen overlay: 33 landmarks and the skeleton connections (MediaPipe default style), a green bbox around the body, and a purple solid circle (tracking point) at the body center.

PD pan-tilt control (tracking + body present only)

Fixed-rate 20 Hz PD control: body center pixel coordinates → normalized error → PD step → limiting (± `max_ptz_step` = 1.2°) → step smoothing (α = `ptz_step_smooth_alpha` = 0.35) → direction reversal (horizontal not reversed, tilt reversed) → angle smoothing (α = `ptz_smooth_alpha` = 0.85) → clamp the angle and publish. When there is no body, no control is output, but the state is not changed and the PD history is not reset — tracking resumes seamlessly once the body is re-detected.

Shortcut keys

Key	Function	Description
Space	idle ↔ tracking	Toggles between standby and tracking
i / I	Back to idle	Stops tracking and pan-tilt control
q / Q / ESC	Exit	Force-exits the node process with `os._exit(0)`

On-screen overlay: The debug window shows the state (idle/tracking), frame rate, body detection status (YES/NO), tracking error H/V, PD parameters, pan-tilt angle, and control rate. A crosshair is drawn at the frame center + an orange deadband rectangle.

Safety and Edge Cases

- No body, no control: when no body is detected, the PD outputs nothing, but the state is not reset; tracking resumes seamlessly once a body is re-detected
- Pan-tilt angle limiting: horizontal [-90°, 90°], tilt [-90°, 20°]
- Deadband protection: the step is forced to zero when the body center offset is within the deadband (20 px)
- Initial PTZ publish at startup: the initial angle (0°, -45°) is published 20 times at 0.2 s intervals
- Exit strategy: press q/ESC to exit directly with `os._exit(0)`
- Runtime parameter validation: full range validation for all PD/image/detection parameters

5. How to Stop

Press `Ctrl+C` in the two terminals to stop each process; it is recommended to close them in the reverse order:

```
# Terminal 2: press Ctrl+C to stop the pose tracking node
# Terminal 1: press Ctrl+C to stop micro_ros_agent
```

4. Experiment Observations

- **idle state**: The frame streams in real time; pose detection works normally but no pan-tilt control is output, and the state is `idle`
- **tracking + body present**: The pan-tilt follows the body center with PD control; the 33-landmark skeleton + green bbox + purple tracking point are overlaid, and the frame shows "Pose:YES" and the error information
- **tracking + body moved out of frame**: The frame shows "Pose:NO", the pan-tilt stays at its last position, and tracking resumes automatically when the body reappears
- **Body within the deadband**: The step is zeroed and the pan-tilt stays completely still without jitter
- **Some landmarks occluded**: The body center is computed by averaging all detectable landmarks, so a reasonable position estimate is still maintained
- **Multiple bodies**: MediaPipe Pose takes the main body in the frame by default and tracks its center
- **Press i to return to idle**: The pan-tilt stays at its current position; press Space to re-enter tracking

5. Program Analysis

> **Source code paths:**
>
> - Launch file: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/pose_tracker_ptz.launch.py`
> - Node file: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/pose_tracker_ptz_node.py`
> - Pose detector: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/pose_detector.py`

1. Core Node Analysis

Node initialization (`__init__`) — create subscribers, publishers, the PoseDetector instance, and dual timers

```
class PoseTrackerPtzNode(Node):

    def __init__(self) -> None:
        super().__init__(DEFAULT_NODE_NAME)
        self._image_lock = threading.Lock()
        self._pose_lock = threading.Lock()

        self._latest_frame: Optional[np.ndarray] = None
```

```
        self._state = STATE_IDLE          # "idle" ↔ "tracking"
        self._has_pose = False
        self._target_x: Optional[float] = None
        self._target_y: Optional[float] = None
        self._h_deg = 0.0                 # PTZ horizontal angle
        self._p_deg = 0.0                 # PTZ tilt angle

        # PD history
        self._prev_err_x = 0.0
        self._prev_err_y = 0.0
        self._prev_ctrl_t = 0.0
        self._last_sx = 0.0
        self._last_sy = 0.0

        self._declare_params()            # declare all ROS2 parameters
        self._load_params()               # load parameters into instance
variables
        self._h_deg = float(self._init_h_deg)
        self._p_deg = float(self._init_p_deg)

        # Initialize the MediaPipe Pose detector
        self._pose_det = PoseDetector(
            detectionCon=float(self._pose_conf),
            trackCon=float(self._pose_track_conf),
        )

        # Create the image subscription (BEST_EFFORT QoS)
        self._img_sub = self.create_subscription(
            Image, self._img_topic, self._on_image,
            QoSProfile(history=QoSHistoryPolicy.KEEP_LAST, depth=1,
                       reliability=QoSReliabilityPolicy.BEST_EFFORT,
                       durability=QoSDurabilityPolicy.VOLATILE),
        )
        # Create the PTZ angle publisher
        self._ptz_pub = self.create_publisher(Vector3, self._ptz_topic, 10)

        # Image processing timer (15 Hz)
        self._img_timer = self.create_timer(
            1.0 / max(float(self._img_rate), MINIMUM_TIMER_RATE_HZ),
            self._process_image,
        )
        # Pan-tilt control timer (20 Hz)
        self._ctrl_timer = None
        self._rebuild_ctrl_timer(False)

        # Initial PTZ angle publishing
        if self._pub_init_ptz:
            self._start_init_ptz()

        self.get_logger().info(
            f"Pose tracker PTZ started. SPACE=toggle, i=idle, q=quit. "
            f"ctrl={self._ctrl_rate:.0f}Hz img={self._img_rate:.0f}Hz"
        )
```

Human pose detection (`_detect`) — 33-landmark average center computation

```
def _detect(self, frame: np.ndarray) -> None:
```

```
    # Call PoseDetector to detect the human pose, returning the drawn frame and
the landmark list
    frame, pose_points, _ = self._pose_det.pubPosePoint(frame)
    if not pose_points:
        with self._pose_lock:
            self._has_pose = False
        return

    # Compute the mean x/y of all 33 landmarks as the body center
    xs = [float(p[1]) for p in pose_points]
    ys = [float(p[2]) for p in pose_points]
    cx = sum(xs) / len(xs)
    cy = sum(ys) / len(ys)

    # Draw the purple tracking point + green bbox
    cv.circle(frame, (int(cx), int(cy)), 8, (255, 0, 255), -1)
    cv.circle(frame, (int(cx), int(cy)), 10, (255, 255, 0), 2)
    cv.rectangle(frame,
                 (int(min(xs)), int(min(ys))),
                 (int(max(xs)), int(max(ys))), (0, 255, 0), 2)

    with self._pose_lock:
        self._has_pose = True
        self._target_x = float(cx)
        self._target_y = float(cy)
```

PD pan-tilt control (`_ctrl_tick`) — normalized error → PD step → smoothing → clamped publish

```
def _ctrl_tick(self) -> None:
    if self._state != STATE_TRACKING:
        return
    with self._pose_lock:
        tx = self._target_x
        ty = self._target_y
        has_pose = self._has_pose
    if not has_pose or tx is None or ty is None:
        return

    # Compute the pixel offset
    icx = float(self._proc_w) * 0.5
    icy = float(self._proc_h) * 0.5
    pex = float(tx - icx)
    pey = float(ty - icy)

    # Deadband handling: force to zero when the offset is within the deadband
    cx = abs(pex) < self._db_x
    cy = abs(pey) < self._db_y
    if cx: pex = 0.0
    if cy: pey = 0.0

    # Compute dt
    n = time.perf_counter()
    dt = n - self._prev_ctrl_t if self._prev_ctrl_t > 0.0 else 1.0 / max(...)
    dt = max(dt, 1e-3)

    # Normalization + derivative
```

```
    nex = pex / icx
    ney = pey / icy
    dx = (nex - self._prev_err_x) / dt
    dy = (ney - self._prev_err_y) / dt

    # PD step computation + limiting
    rx = clamp((self._kp * nex + self._kd * dx) * self._max_step,
               -self._max_step, self._max_step)
    ry = clamp((self._kp * ney + self._kd * dy) * self._max_step,
               -self._max_step, self._max_step)

    # Step EMA smoothing
    a = self._step_alpha
    sx = (1.0 - a) * self._last_sx + a * rx
    sy = (1.0 - a) * self._last_sy + a * ry
    if cx: sx = 0.0; nex = 0.0
    if cy: sy = 0.0; ney = 0.0

    self._prev_err_x = nex
    self._prev_err_y = ney
    self._prev_ctrl_t = n
    self._last_sx = sx
    self._last_sy = sy

    # Direction reversal
    if self._rev_h: sx = -sx
    if self._rev_p: sy = -sy

    # Angle smoothing + limiting + publish
    sa = self._ptz_alpha
    self._h_deg = clamp((1.0 - sa) * self._h_deg + sa * clamp(self._h_deg + sx,
...), ...)
    self._p_deg = clamp((1.0 - sa) * self._p_deg + sa * clamp(self._p_deg + sy,
...), ...)
    self._pub_ptz()
```

2. Key Parameters

> **Parameter definition file:**
> `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/pose_tracker_ptz.launch.py`

Parameter	Type	Default Value	Description
`pd_kp` / `pd_kd`	float	`0.7` / `0.25`	PD proportional/derivative gains
`control_rate_hz`	float	`20.0` Hz	Pan-tilt control rate
`image_processing_rate_hz`	float	`15.0` Hz	Image processing rate
`deadband_x` / `deadband_y`	float	`20.0` / `20.0` px	Horizontal/vertical deadband
`max_ptz_step`	float	`1.2` °	Maximum single angle step
`ptz_step_smooth_alpha`	float	`0.35`	Step EMA smoothing coefficient
`ptz_smooth_alpha`	float	`0.85`	Pan-tilt angle EMA smoothing coefficient
`pose_detection_confidence`	float	`0.5`	Pose detection confidence threshold
`pose_tracking_confidence`	float	`0.5`	Pose tracking confidence threshold
`processing_width` / `processing_height`	int	`320` / `240` px	Processing resolution
`display_width` / `display_height`	int	`960` / `720` px	Debug display resolution
`initial_horizontal_deg` / `initial_pitch_deg`	float	`0.0` ° / `-45.0` °	Initial pan-tilt angles

6. Frequently Asked Questions

Problem	Cause	Solution
Obvious lag	Pose inference (33 landmarks) is computationally heavy + the 15 Hz image processing adds double delay	Lower the processing resolution (e.g., 240×180) or accept some tracking delay
Tracking center offset	Some landmarks (e.g., limb tips) are far from the body center, skewing the mean	This is normal — the mean center reflects the overall distribution; for extreme offsets, run `ros2 param set /pose_tracker_ptz_node deadband_x 30.0` and `ros2 param set /pose_tracker_ptz_node deadband_y 30.0` to adjust the deadband
No detection after startup	The MediaPipe environment script was not sourced	First run `source /opt/ros/humble/setup.bash` `source ~/microros_ws/install/setup.bash`
Body lost when partially out of frame	Fewer landmarks make the mean jump	Keep the body in the center region of the frame; the pan-tilt tracking auto-centers it